t3-w05 — Three dimensional shapes II — Three dimensional figures: properties of cylinders and spheres, and volume

Run `diag-v3` · conditions C0 and C1 · the pack shown is draw 3 of each.

C0 `The curriculum objective alone. No lesson plan, no note.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three dimensional shapes II

`subtopic`

Three dimensional figures: properties of cylinders and spheres, and volume

`term`

Third Term

`week`

5

`lesson_objectives`

• identify properties of cylinders and spheres
• find the volume of a cube and a cuboid

Objective 1 — identify properties of cylinders and spheres

`objective_text`

identify properties of cylinders and spheres

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Objective 2 — find the volume of a cube and a cuboid

`objective_text`

find the volume of a cube and a cuboid

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

Properties of Cylinders and Spheres

> EXPLANATION
> **Cylinder**
> - Two identical circular faces called bases (flat)
> - A curved lateral surface connecting the bases (curved)
> - Height: the distance between the bases
> - Radius: distance from the centre of a base to its edge
>
> **Sphere**
> - All points on the surface are the same distance from the centre (radius)
> - No edges, vertices, or flat faces
> - Completely smooth and round

> WORKED_EXAMPLE
> **Worked Example**
> Consider a soup can. It has two flat circular ends (the top and bottom) and a curved side that wraps around. These match the description of a cylinder, so the can is a cylinder.
>
> Now think of a basketball. It has no flat parts; every point on its surface is the same distance from the centre. This matches the description of a sphere, so the basketball is a sphere.

1. List the parts of a cylinder and indicate which are flat and which are curved.
answer: Two circular bases – flat; one curved lateral surface – curved.

2. State three key properties of a sphere.
answer: All points on the surface are the same distance (radius) from the centre; it has no edges, vertices, or flat faces; it is perfectly round.

3. A solid has two identical circular ends and a curved side connecting them. Is it a cylinder or a sphere? Explain your answer.
answer: It is a cylinder because it has two flat circular ends (bases) and a curved lateral surface; a sphere would have no flat ends.

Finding the Volume of Cubes and Cuboids

> EXPLANATION
> The volume of a solid tells us how much space it occupies.
>
> **Cube**: All three edges are the same length. If the length of each edge is \(s\) centimetres, the volume is
>
> \[V = s \times s \times s = s^3 \]
>
> **Cuboid (rectangular prism)**: It has three different edge lengths – length \(l\), width \(w\) and height \(h\). Its volume is
>
> \[V = l \times w \times h \]
>
> Remember to keep the units consistent (e.g., all in centimetres) and the final answer will be in cubic centimetres (cm³).

> WORKED_EXAMPLE
> ### Example 1 – Cube
> A cube has a side length of 4 cm.
>
> Volume \(= 4 \times 4 \times 4 = 64\) cm³.
>
> ### Example 2 – Cuboid
> A cuboid measures 5 cm by 3 cm by 2 cm.
>
> Volume \(= 5 \times 3 \times 2 = 30\) cm³.

1. A cube has a side length of 6 cm. What is its volume?
answer: 216 cm³

2. A cuboid measures 7 cm by 4 cm by 3 cm. Find its volume.
answer: 84 cm³

3. The volume of a cube is 125 cm³. What is the length of each side?
answer: 5 cm

4. A rectangular box (cuboid) has a volume of 240 cm³ and its base measures 8 cm by 5 cm. Find its height.
answer: 6 cm

C1 `The objective plus the lesson plan's design: position, method labels, materials, entry behaviour and assessment.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three dimensional shapes II

`subtopic`

Three dimensional figures: properties of cylinders and spheres, and volume

`term`

Third Term

`week`

5

`lesson_objectives`

• identify properties of cylinders and spheres
• find the volume of a cube and a cuboid

`entry_behaviour`

• properties of cubes and cuboid
• properties of pyramids and cones

`instructional_materials`

• shapes of cylinders and spheres.

Objective 1 — identify properties of cylinders and spheres

`objective_text`

identify properties of cylinders and spheres

`current_step`

1

`instructional_events`

• present_content
• elicit_performance
• present_content

`student_tasks`

• 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 2 — find the volume of a cube and a cuboid

`objective_text`

find the volume of a cube and a cuboid

`current_step`

2

`instructional_events`

• present_content
• provide_learning_guidance

`assessment_points`

• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high.
• Calculate the volume of cuboid with dimensions 5m by 2m by 3m.
• What is the volume of a cube of 4cm $\times$ 4cm $\times$ 4cm.
• A rectangular tank hold 6m long by 2m wide holds 36m^{3} of water. How deep is the water in the tank.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

Properties of Cylinders and Spheres

> EXPLANATION
> A solid cylinder has two circular faces (the top and bottom bases) and one curved surface that wraps around the sides. In geometry we count the curved surface as a face, so a cylinder has a total of three faces. It also has two edges where the circular faces meet the curved surface.
>
> A sphere is different: it has no flat faces or edges. Every point on its surface is the same distance from the centre, so we say a sphere has zero faces and zero edges, but it has one continuous curved surface.

> EXPLANATION
> A sphere is a perfectly round three-dimensional shape. Because it has no flat surfaces, we say it has no faces or edges. Its entire surface is a single curved surface, and every point on that surface is the same distance from the centre.

1. How many circular faces does a solid cylinder have?
answer: 2

2. A cylinder has how many faces?
answer: 3

3. How many faces does a sphere have?
answer: 0

Objective 2

Finding the Volume of Cubes and Cuboids

> EXPLANATION
> The volume of a three-dimensional shape tells us how much space it occupies. For any rectangular solid (a cuboid) you find the volume by multiplying its length, width and height:
>
> $$ V = \text{length} \times \text{width} \times \text{height} $$
>
> All three dimensions are measured in the same units, and the volume is expressed in cubic units (e.g., cm^3, m^3).
>
> A cube is a special type of cuboid where all three edges are the same length, say s. Its volume formula becomes:
>
> $$ V = s \times s \times s = s^3 $$
>
> Remember to keep the units consistent and to write the final answer with the appropriate cubic unit.

> WORKED_EXAMPLE
> **Worked Example:** Find the volume of a cuboid that is 2 cm long, 3 cm wide and 4 cm high.
>
> 1. Write down the dimensions: length = 2 cm, width = 3 cm, height = 4 cm.
> 2. Multiply them together:
> $$ V = 2 \times 3 \times 4 = 6 \times 4 = 24 $$
> 3. State the answer with units:
> $$ V = 24\ \text{cm}^3 $$
>
> So the cuboid occupies 24 cubic centimetres of space.

1. Calculate the volume of a cuboid 8 cm long, 4 cm wide and 3 cm high.
answer: 96 cm^3

2. Calculate the volume of a cuboid with dimensions 5 m by 2 m by 3 m.
answer: 30 m^3

3. What is the volume of a cube of 4 cm × 4 cm × 4 cm?
answer: 64 cm^3

4. A rectangular tank 6 m long by 2 m wide holds 36 m^3 of water. How deep is the water in the tank?
answer: 3 m

Result — per rubric dimension

Scores are 1–5, each the mean of 2 judge pass(es) on the same packet. The first three columns are the packs shown above (draw 3); the last is the lesson's mean over its 3 draws per condition — the study's number.

Rubric	C0	C1	C1 – C0, shown packs	C1 – C0, mean of 3 draws
Objective and curriculum alignment	5.00	5.00	+0.00	**+0.17**
Taught-lesson alignment	4.00	4.00	+0.00	**+0.17**
Scope and conceptual correctness	5.00	4.00	**-1.00**	**-0.17**
Student-facing clarity and usefulness	5.00	4.50	**-0.50**	+0.00